ECE3232/ME4673

Competition Guide

W2026

Nov 4 2026 (Rev 1.2)

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INTRODUCTION

Overview

Welcome to The Harvest! A joint initiative between ECE3232 and ME4673 to improve your engineering design and technical skills all while having fun! This innovative competition will serve to meet the project-based requirements of your course and allow for innovation above and beyond expectations.

Purpose of the Manual

This manual will be your guide throughout the term. Get to know this manual and the rules set within as you can consider this your requirements for your design.

Terminology

Throughout this manual, various terms may be used that require further clarification.

Rover: A rover is a small, wirelessly controlled, robot that is designed by groups in ECE3232. The rovers serve to unlock challenges and ore on the map for the harvesters to harvest.

Harvester: A harvester is a larger wirelessly controlled robot that is designed by groups in ME4673. The harvesters serve to collect ore on the playing field and from challenges unlocked by rovers. Harvesters also serve to bring ore back to their home base processing plant for further processing.

Processing Plant/Home Base: Each team will be provided with a processing plant that serves as their home base. The processing plant will be where harvesters return ore to for sorting and additional points.

Group: A group is course specific and consists of your assigned 2-4 members. I.e. all members in your group are in the same class as you. Each group is responsible for one rover or harvester, depending on which course you are registered in (ECE3232 or ME4673).

Team: A team is made up of multiple groups from both courses. A team will consist of at least one or more harvesters and one or more rovers and will compete head-to-head against another team during the competition.

RCLS: The Radio Controlled Laser System is a custom designed PCB that is connected to a mikroBus on your Peripheral Controller Unit (PCU). The RCLS interfaces with your laser, laser sensor, and radio-controlled wireless receiver.

RCS: The Rover Control System is comprised of an MPLABXpress development board connected to a Peripheral Control Unit (PCU)

HCS: The Harvester Control System is comprised of a UNB Dev Board connected to a Peripheral Control Unit (PCU).

PCU: The Peripheral Control Unit is a UNB Dev Board in disguise with custom firmware. This UNB Dev board will have firmware provided by your course instructors that will handle access to peripherals such as the motor drivers and RCLS for laser system control. Additionally, your PCU will communicate (automatically) with

the CCS through the WCU to report points, health, damage and any other competition related information.

WCU: The Wireless Control Unit is a temporary solution to add wireless communication to the PCU, which was found to have a faulty Bluetooth antenna. The WCU will be a BBC micro:bit running custom code that will be provided to the student. Instructions on how to connect the WCU to your PCU will be provided

CCS: The Competition Control System will be responsible for tracking all rover and harvester points, unlocking gates and any other competition-related tasks. Students will not directly interact with the CCS.

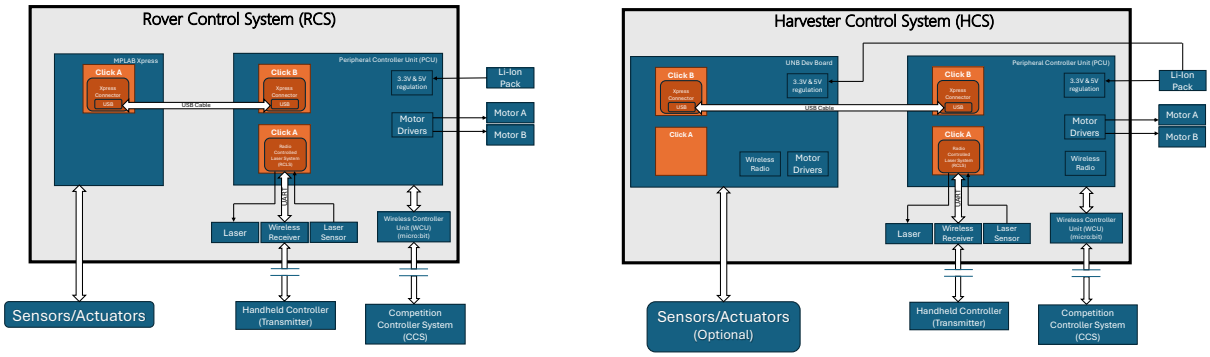


Figure 1 RCS and HCS Block Diagrams

ROVER & HARVESTER CONSTRUCTION INFORMATION

Overview

When designing and constructing your rover or harvester, there are often questions about what materials can/can not be used. Our general guideline is that it's **best to check with your course instructors** for anything that deviates from the standard kit.

A 3D printable rover frame (STL) is available for **ECE3232** students to use for their project, but take note that it is not required with custom designs often encouraged. **ME4673** students must design all frame aspects from scratch.

COMPETITION OVERVIEW

Description of the Playing Field

The playing field will be a section of floor within a space in Head Hall. The playing field will be made up of modular tasks as described in this document. The placement of the modular tasks will be classified until

competition day, but the specifics of each task can be found in this document. The floor surface between modular tasks will be tiled and obstruction free.

Generic Rules/Guidelines/Tips

- No harvester or rover may intentionally cause damage to another harvester or rover.
- No harvester or rover may directly or indirectly block another harvester or rover.
- A harvester or rover may choose to be reset to their respective base at any time it becomes stuck or non-responsive for a period of 10 seconds or more. If the harvester or rover is preventing the movement or operation of another harvester or rover after this time period, it will automatically be moved to its respective home base.
- Aside from the roles/constraints listed below, harvesters and rovers may choose to overlap in functionality.
- Once ore has reached an opposing team's base, it may **NOT** be taken.
- Ore on a Harvester or Rover may **NOT** be taken by another team.
- All ore on the map is up for grabs regardless of who unlocked a gate. There's no 5 second rule when it comes to dropping your ore either!

Roles of Harvester and Rovers

To ensure that your design meets the course requirements, the following design constraints for a harvester and rover are outlined below:

Harvester

- Harvesters must be mobile.
- Harvesters must auto sort all incoming ore according to type (colour).
- Harvesters must be able to dispense a selected ore type all at once into their processing plant.
- Harvesters have no limit on the number of ore that can be sorted and stored.
- Upon **start** of the round, harvesters are limited to a **30cm x 30cm** footprint with no restrictions on height.

Rover

- Rovers must be mobile.
- Rovers may only store/manipulate/push **1 piece** of ore at a time.
- Each **group** must design to accomplish a minimum of **1 unique task** from Table 1 **per member** of their group.
- Redundancy of tasks IS allowed in a Team. I.e. Multiple rovers within a team may design to accomplish the same tasks but it is recommended to spread out the tasks for maximum points.
- Upon **start** of the round, rovers are limited to a **20cm x 20cm** footprint with no restrictions on height.

SCORING

Overview

Points can be achieved in a variety of different ways to ensure variation in strategies and to prevent the failure of any one rover or harvester from causing a guaranteed loss of the competition.

Quick Point Summary

A quick view table is provided to allow you to sort through how points are awarded. For further information, refer to the detailed point summary section below. Note that all tasks with points will unlock 5 pieces of ore.

Task Name	Task Description	Points
Base Stockpile	Ore at base at end of round	1
Unit Stockpile	Ore on Harvester or rover at end	*2
Processing Stockpile	Ore in processing bins at end of round	*4
RFID Sequence	RFID tag read and beamed to home base	10
Optical Signal Decoding	Convert colours to tones to unlock ore	10
Environmental Remediation Task	Drain tailings pond to unlock ore gate	10
Solar Array	Charge solar array to unlock ore gate	5
Laser Turret	Activate Harvester Shields to provide safe passage to ore	10
Magnetic Anomaly	Detect magnetic fields to unlock ore gate	5
Autonomous Line	Autonomous line follow to unlock ore gate	15
Alien Frequency	Find the fundamental to unlock ore gate	20
Electrical Conductivity	Detect conductive material to unlock ore gate	10
IR Detection	Detect IR emission to unlock ore gate	10
Ore	Open area with 10 pieces of ore and no associated task	0

Table 1 - Point Overview

***Note:** *Penalties described in detailed section below may alter points*

Detailed Point Summary

For every task given in Table 1, a detailed overview is provided.

Base Stockpile

Overview:

- The "Base Stockpile" task involves collecting ore spread across the playing field and bringing it back to the team's home base. This task is fundamental and open to all rovers and the harvester.

Technical Requirements:

- None

Procedure:

- Transport ore back to your team's home base.

Scoring Criteria:

- Each piece of ore successfully brought back to the home base and within base boundaries at the end of the round is worth 1 point.
- **Note:** Ore on rovers or the harvester at the end of the round does not count towards this task.

Rules and Constraints:

- Ore must be physically within the home base boundaries at the end of the round.
- When viewed from top down, the ore must be 'touching' the base boundaries line or within.
- Removing ore from the opposing team's base is **NOT** permitted.

Tips and Strategies:

- Consider the balance between a rover transporting one piece of ore to a base vs transporting to a harvester (which can take more than one piece at a time)
- Remember that ore at the base is the least number of points possible and is a last resort. Ore found on the harvester or rovers at the end of time or in the processing plant is worth more.

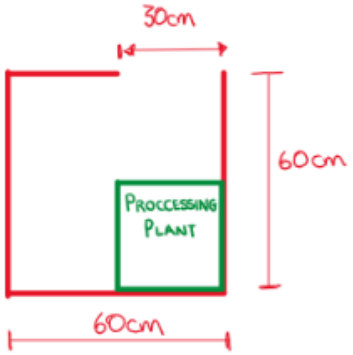


Figure 2 Home Base/Processing Plant

Unit Stockpile

Overview:

- This task involves both the harvesters and rovers storing ore on themselves. The harvester must sort the ore by color, while rovers can only carry one piece of ore at a time.

Technical Requirements:

- **Harvesters**: Color sensor for ore detection and automated sorting mechanism.
- **Rovers**: Basic carrying capacity, limited to one ore at a time.

Procedure:

- Harvesters and rovers collect ore from the playing field.
- Harvesters uses a color sensor to detect and **automatically sort** the ore into separate compartments.
- Rovers focus on transporting individual ore pieces only.

Scoring Criteria:

- Points awarded for each piece of ore on a harvester or a rover at the end of the round.
- For the harvester, mixed (unsorted) ore counts for half points.

Rules and Constraints:

- Ore must be on a harvester or a rover at the end of the round to count.
- Harvesters must sort the ore by color for full points.
- Rovers can carry/push/pull only one piece of ore at any given time.

Tips and Strategies:

- Develop an efficient and accurate automated sorting system on the harvester to maximize point potential.
- Consider using rovers to bring ore to store on harvester instead of rover for processing potential.
- Consider dumping mechanism on harvester if ore becomes mixed in order to harvest again and maximize scoring potential.
- Remember that processing ore is more valuable than storing, so aim to process and not just store.

Processing Stockpile

Overview:

- In the "Processing Stockpile" task, the harvester robot is responsible for depositing ore into a processing plant. The type of ore to be deposited at any given time must be communicated to the processing plant via a rover. The processing plant will provide a display to indicate the type of ore it is ready to process.

Technical Requirements:

- **Harvester:** Mechanisms for retrieving and depositing ore into the processing plant.
- **Rover:** Communication system for communicating with the processing plant.

Procedure:

- The harvester collects and sorts ore, then transports it to the processing plant.
- Rover(s) communicate with the processing plant to indicate which type of ore the harvester is ready to deposit
- The harvester deposits as many of the indicated ore types as it likes.

Scoring Criteria:

- Properly sorted ore in the processing plant gains full points per ore at end of round.
- Improperly sorted ore (mixed in a bin) will reduce all contaminated ore by **half points**.

Rules and Constraints:

- Ore must be sorted correctly in the processing plant for full points.
- Rovers must accurately communicate with the processing plant to ensure the correct ore type is processed.

Tips and Strategies:

- Coordinate between the harvester and rovers to ensure accurate communication and efficient ore deposit.
- Develop a reliable system on the rover for communication with the processing plant.

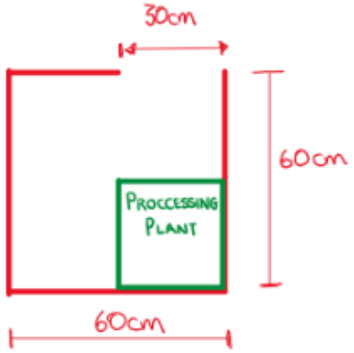


Figure 3 Home Base/Processing Plant

RFID Gate

Overview:

- The "RFID Gate" challenge involves a gated area on the map protecting ore, with an embedded RFID tag in the gate. Rovers are required to read this RFID tag and then transmit the first 8 digits of the tag ID to your home base using your built-in communication system. A correct transmission of the ID will unlock the gate, granting access to the ore for the harvester and earning the team points.

Technical Requirements:

- **Rover**: RFID reader to scan the tag.
- **Communication System**: A way to transmit the RFID read from the tag in the gate back to the home base.

Components Provided:

RFID Reader
DigiKey: 4411-CN0090-ND

Procedure:

- Navigate the rover to the RFID gate.
- Use the RFID reader to scan the tag and obtain the ID.
- Transmit the RFID tag ID from the rover to home base using your communication system.
- The processing plant receives the code and sends a signal to unlock the gate if the ID is correct.

Scoring Criteria:

- Successfully unlocking the gate gains points
- The task also enables the harvester to access and collect the protected ore.

Rules and Constraints:

- The gate will only unlock if the correct RFID tag ID is transmitted.
- Incorrect transmissions will not unlock the gate and will not yield points.

Tips and Strategies:

- Coordinate with the harvester to be ready to collect the ore as soon as the gate is unlocked to avoid the other team stealing them!

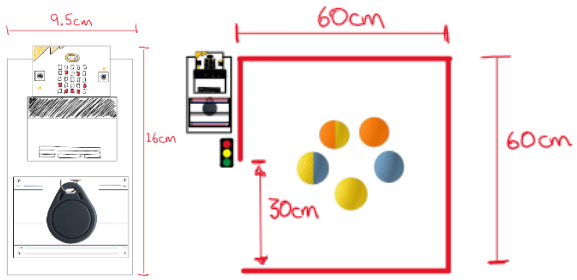


Figure 4 RFID system and tag

Optical Signal Decoding Challenge

Overview

- In the "Optical Signal Decoding Challenge," the rover is tasked with interpreting a sequence of RGB signals emitted from a signal station, representing encoded communication from an alien planet. The rover must then convert these optical signals into an audio tone sequence and play them back, demonstrating successful translation and transmission of interstellar messages to unlock an area of ore for the harvester.

Technical Requirements

- **Rover**: Equipped with a light sensor to detect light signals and a speaker or tone generator to convert and replay the signals as audio tones.

Components Provided:

Colour Sensor	Speaker
DigiKey: 1528-2395-ND	Mouser: 713-107020109

Procedure

- The rover navigates to the signal station on the map.
- The station emits a sequence of colour signals in a continual pattern.
- The rover uses its light sensor to detect and translate the colours to tones in **real time**. Tones must match the colour shown and remain on while the colour is shown.

Scoring Criteria

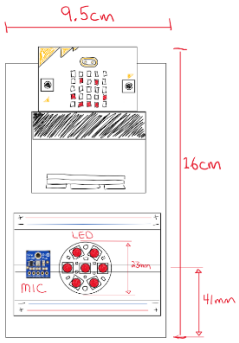
- Accurate interpretation and playback of the optical-to-audio signal sequence are awarded points.
- Additional points available by harvesting the ore that is unlocked.

Rules and Constraints

- The rover must accurately detect and interpret the optical signals and convert them to the correct audio tones.
- The rover must do the detection, conversion and playback sequence automatically. That is to say, the colour to tone map provided below must be hard coded into the device. The colour sequence will be random and vary per each round (to prevent pre-coding).
- Red, Green and Blue patterns will be the only colours shown.

Tips and Strategies

- Implement and calibrate sensitive light detection on the rover to accurately capture colour variations.
- Test the accuracy of your tone production frequency (use a phone app).



Colour	Note
Red	C4
Green	F4
Blue	A4

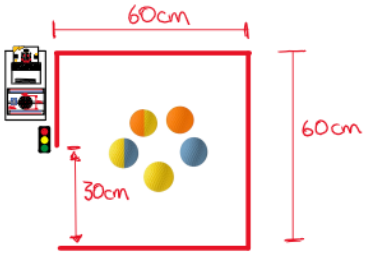


Figure 5 Gate System with LED and Microphone and Colour/Note Table

Environmental Remediation Task

Overview

- The "Environmental Remediation Task" simulates a scenario on a foreign planet where a rover encounters a mining tailings pond. The challenge involves safely relocating the water from the pond to your designated containment area. Whichever team fills their containment area first gets the points and unlocks more ore.

Technical Requirements

- **Rover**: Equipped with a water pump, motor driver and mechanism to direct water to teams area.

Components Provided:

Water Pump	Motor Driver
Amazon	**DigiKey**:1528-1212-ND

Procedure

- Navigate the rover to the tailings pond area.
- Use the onboard pump to extract water from the pond and direct it to your teams containment area, ensuring it does not spill or leak (you may not carry the water).
- Once enough water is relocated (water level sensor), a gated area with ore is unlocked.

Scoring Criteria

- Points are awarded based on reaching the min height of remediated water in your containment area.
- Additional points are granted for unlocking the gated ore area.

Rules and Constraints

- The gated area only unlocks if a predetermined amount of water is successfully relocated into **YOUR** remediation pond.

Tips and Strategies

- Ensure pump intake can be directed into tailings pond and outlet directed to your containment area.
- Monitor the water levels to strategically plan when to relocate water for unlocking the gated area.
- If the other Team also takes enough water out of the pond, it may not leave enough to trigger your containment area water level sensor, so act quickly!

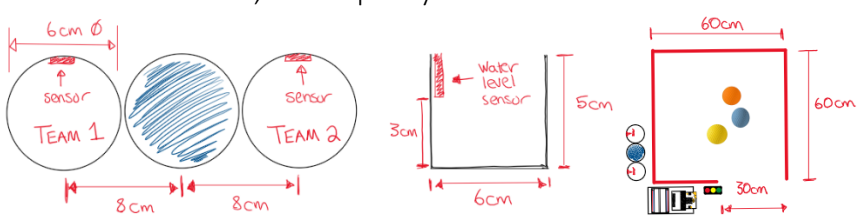


Figure 6 Environmental Remediation Layout

Solar Array Activation Mission

Overview:

- In the "Solar Array Activation Mission," teams encounter a scenario where a gate protecting valuable ore reserves is powered by a solar array. Teams must direct sufficient light to a solar panel to charge it. Once charged, the gate opens, allowing access to the ore. The gate remains open as long as the panel is sufficiently charged but begins to close as the charge depletes.

Technical Requirements:

- **Rover**: Equipped with a light source strong enough to charge the solar panel. **(Must use driving circuit)**

Components Provided:

LED	Driving MOSFET
DigiKey: C512A-WNN-CZ0B0151-ND	**DigiKey: 2N7000FS-ND**

Procedure:

- Navigate the rover to the location of the solar panel.
- Use the rover's light source to direct light onto the panel, charging it.
- Maintain the light on the panel until the gate opens, revealing the ore.
- Collect the ore before the gate closes as the panel starts to lose its charge.

Scoring Criteria:

- Points will be awarded for successfully opening the solar gate.
- Points are awarded based on the amount of ore collected from behind the gate.

Rules and Constraints:

- The gate will only stay open while the panel remains charged.
- A LED driving circuit must be used (not just simple I/O)
- The LED must be controllable to turn on/off

Tips and Strategies:

- Optimize the position and angle of the rover's light source for maximum charging efficiency.
- Plan the timing of the charging process and ore collection to maximize the duration the gate stays open.

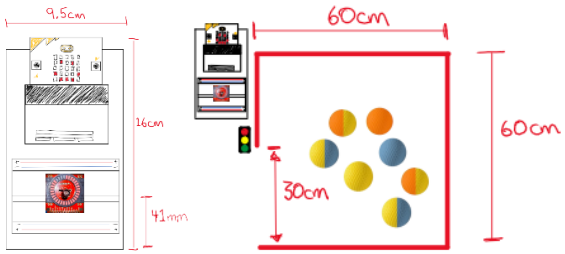


Figure 7 Solar Array Layout

Laser Turret Defense Challenge

Overview

- In the "Laser Turret Defense Challenge," teams face a unique scenario where a valuable ore section is guarded by laser turrets. These turrets are equipped with pulse-coded lasers that can disable any rover or harvester for a set period of time if hit. This laser hit carries encoded information which a rover can decode. Once decoded, the rover may use its own pulse-coded laser system to transmit a 'shield code' to any other rover or harvester, protecting them from being disabled by the turrets.

Technical Requirements:

- **Rover**: Equipped with RCLS and laser sensor (mandatory)

Components Provided:

None

Procedure:

- A rover navigates towards the laser turrets and allows itself to be hit by the turret's laser.
- The rover decodes the information contained in the laser pulse.
- After its disabled period is over, it transmits the shield code to a harvester using the RCLS
- The harvester can now safely access the ore without being disabled by the turrets.

Scoring Criteria:

- Points are awarded for successfully decoding the turret's laser signal and transmitting the shield code to the harvester.
- Additional points are granted for each piece of ore safely collected from the protected area.

Rules and Constraints:

- **All systems on the rover** will be temporarily **disabled** upon receiving the turret's laser hit.
- The harvester must receive the correct shield code to avoid being disabled by the turrets.

Tips and Strategies:

- Volunteer a rover as tribute to take a hit and be disabled such that it can learn the shield code and transmit to the harvester after the disabling period is over.
- It is not recommended to send an unprotected Harvester in, but it IS permitted.

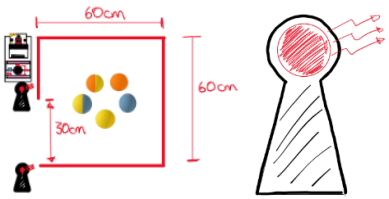


Figure 8 Laser Turret System

Magnetic Anomaly Detection Challenge

Overview:

- "Magnetic Anomaly Detection Challenge" involves navigating a planet scattered with identically appearing, numbered pyramids. Hidden within one pyramid is a magnet, symbolizing a mysterious magnetic anomaly. Teams use a Hall effect sensor to determine the magnet's location. A rover that uses it's pulse-coded laser to beam the correct identification back unlocks a new ore deposit on the field, while incorrect guesses result in point deduction.

Technical Requirements:

- **Rover**: Equipped with a Hall effect sensor for magnetic field detection and pulse coded laser.
- **Pyramids**: Identical in appearance, each uniquely numbered, with one containing a hidden magnet.
- **PCU**: A way for the rover to transmit the identified magnetic anomaly to the processing plant.

Components Provided:

Hall Effect Sensor
DigiKey: 296-DRV5053CAQLPGMCT-ND

Procedure:

- Rovers scan each numbered pyramid to detect the presence of a magnetic field.
- Once a team believes they have identified the correct pyramid, they transmit its number back to their processing plant.

Scoring Criteria:

- Points are **awarded** for correctly identifying the number of the pyramid containing a magnet.
- Points are **subtracted** for incorrectly identifying a pyramid.

Rules and Constraints:

- While users are allowed to make guesses without performing the task, the loss of points will far outweigh the benefit of eventually getting a correct guess.

Tips and Strategies:

- Ensure the hall effect sensor is properly calibrated and sensitive enough to detect magnetic fields.
- Quickly and accurately communicate the findings to avoid another team stealing the points.

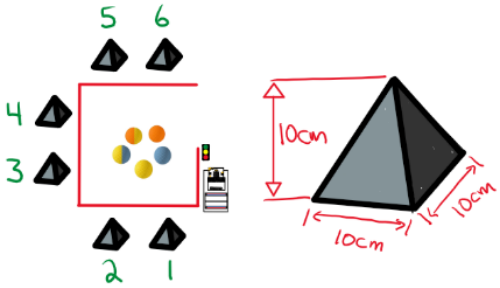


Figure 9 Magnetic Anomaly Layout

Autonomous Trail Tracking Challenge

Overview:

- The "Autonomous Trail Tracking Challenge" simulates a scenario where a rover must navigate a special path autonomously, representing the tracking of a unique planetary force or energy trail on an alien landscape. This path is marked by a high contrast line on the floor. To unlock a valuable ore deposit located at the end of this trail, the rover must demonstrate its ability to follow the line independently, without any manual guidance to a judge nearby.

Technical Requirements:

- **Rover**: Equipped with line-following sensors and autonomous navigation capabilities.

Components Provided:

Line Following Array
Robotshop

Procedure:

- Position the rover at the start of the marked line.
- Engage the rover's autonomous navigation system to follow the line.
- Upon reaching the end of the line, the ore will be unlocked and points awarded.
- Teams must show a judge that the rover is navigating autonomously at the start of the trail.

Scoring Criteria:

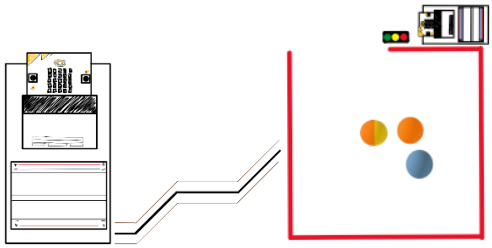
- Successful autonomous navigation of the entire trail results in unlocking the ore and earning points.
- Additional points for retrieval of the unlocked ore by a Harvester.

Rules and Constraints:

- The rover must follow the line autonomously without any external control or intervention.
- The rover must stay on the line throughout the challenge to qualify for points.
- There are no limitations to the number of tries, but failed attempts are reset back to the start.

Tips and Strategies:

- Thoroughly test and fine-tune the rover's line-following algorithm and sensors to ensure reliable performance.
- Optimize the rover’s speed and maneuverability to navigate the trail efficiently.
- Prepare a clear demonstration of the rover's autonomous capabilities for the judge.



- *Figure 10 Example Line Following Path (Not accurate)*

Alien Frequency Analysis Challenge

Overview:

- In the "Alien Frequency Analysis Challenge" teams encounter a unique sound emanating from a specific location on the alien planet. This sound, a blend of white, brown, or pink noise, contains a hidden fundamental frequency. The task for the rovers is to approach the sound source, analyze the noise to isolate the fundamental frequency and then accurately transmit this frequency back to their home base.

Technical Requirements:

- **Rover:** Equipped with microphone and FFT (or equivalent) algorithm programmed
- **PCU**: A way to transmit the identified conductive material number to the processing plant.

Components Provided:

Microphone	Op-Amp
DigiKey: 1528-1362-ND	**DigiKey**: 296-1893-5-ND

Procedure:

- Navigate the rover to the sound source location on the map.
- Record the emitted noise and use FFT(or equivalent) to analyze and identify the fundamental frequency.
- Teams have up to three attempts to correctly identify and transmit the frequency (integer), with an allowance of +/- 10 % Hz back to the home base.

Scoring Criteria:

- Points are awarded for accurately identifying and transmitting the fundamental frequency within the specified tolerance range using the PCU.

Rules and Constraints:

- The rover must be close to the sound source to ensure a clear recording.
- The frequency identification must fall within +/- 10% Hz of the actual fundamental frequency.
- The frequency range will fall between 100Hz-4KHz
- Teams are limited to three attempts to transmit the correct frequency. Further attempts will be ignored.

Tips and Strategies:

- Thoroughly test the FFT implementation for reliability in isolating the fundamental frequency from a noisy background.
- Plan strategically to make each attempt count, especially considering the limited number of tries

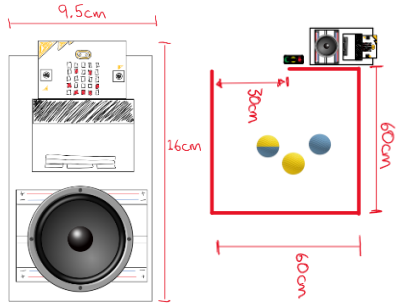


Figure 11 Gate system with embedded frequency generation circuitry

Electrical Conductivity Exploration Challenge

Overview

- The "Electrical Conductivity Exploration Challenge" simulates a task where rovers explore various areas on an alien planet to analyze the electrical conductivity of different substances. These areas, each with a unique number, contain substances with varying levels of conductivity. The rovers are equipped with tools to measure this conductivity. Correctly identifying the only conductive material and reporting it to the processing plant unlocks additional ore and earns points. Incorrect guesses result in point deductions, reflecting the costs of inaccurate scientific analysis.

Technical Requirements

- **Rover**: Equipped with a conductivity sensor or probe to measure the electrical conductivity of substances.
- **Map Areas**: Designated spots on the playing field, each with a different substance and a unique identifying number.
- **PCU**: A way to transmit the identified conductive material number to the processing plant.

Components Provided:

No sensor provided
Up to group to research how to complete with simple objects.

Procedure

- Navigate the rover to the various numbered areas on the map.
- Use the conductivity sensor to measure the electrical conductivity of the substance at each location.
- Determine which area contains the most conductive material.
- Report the number of that area to the processing plant.

Scoring Criteria:

- Points are **awarded** for correctly identifying the identification number of the mystery box with conductive material inside.
- Points are **subtracted** for incorrectly identifying a mystery box (i.e. a mystery box with a non-conductive material).

Rules and Constraints:

- While users are allowed to make guesses without performing the task, the loss of points will far outweigh the benefit of eventually getting a correct guess.

Tips and Strategies:

- Ensure your conductivity system is properly calibrated and sensitive enough to detect the difference between conductive and non-conductive material.
- Quickly and accurately communicate the findings to avoid another team stealing the points.

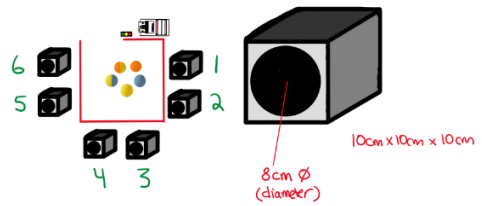


Figure 12 Conductivity Task Layout and Conductive Mystery Boxes

IR Light Detection Mission

Overview:

- The "IR Light Detection Mission" places rovers in a scenario where they must identify active infrared (IR) light sources on an alien planet. The playing field features various numbered locations, each equipped with an IR transmitter. Only one of these IR LEDs are active per round. Rovers are tasked with scanning these locations using IR sensors to identify the active IR light sources. Successfully detecting an active IR transmitter and using the PCU to send the location number back to the home base results in points being awarded and additional ore being unlocked.

Technical Requirements:

- **Rover**: Equipped with a IR light sensor capable of detecting active IR light sources.
- **Map Locations**: Designated spots on the map, each numbered, with potential IR light transmitters.
- **PCU**: A way to transmit the identified active IR light number to the processing plant.

Components Provided:

IR Receiver
DigiKey: TSOP4830-ND

Procedure:

- The rover navigates to one of the numbered IR transmitters.
- Using its IR sensor, the rover attempts to detect if the current IR transmitter is active or not.
- Once an active IR light source is detected, the rover transmits the task info and active number to their processing plant.

Scoring Criteria:

- Points are **awarded** for correctly identifying the number of the IR transmitter that is active.
- Points are **subtracted** for incorrectly identifying the active transmitter.

Rules and Constraints:

- While users are allowed to make guesses without performing the task, the loss of points will far outweigh the benefit of eventually getting a correct guess.

Tips and Strategies:

- Ensure the IR sensor is properly calibrated and sensitive enough to detect IR light accurately.
- Quickly and accurately communicate the findings to avoid another team stealing the points.

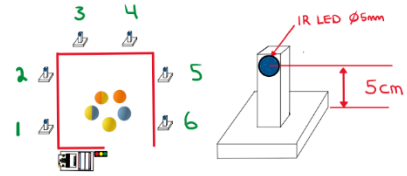


Figure 13 IR Task layout and Transmitter

PULSE-CODED LASER SYSTEM

Overview

- The pulse-coded laser system on the RCLS can be used as an offensive weapon to shoot and temporarily disable a rover or harvester.

- A rover can shoot a rover from another team to overload their onboard power system. This overload will temporarily disable the rover for a specific period which will allow the power system to recuperate from this overload event. A team's rovers may also attempt to shoot the opponent's harvester to disable it. Since the harvester consists of a more robust power system (compared to those of the rovers), a harvester must sustain 3 hits, each from a different rover, before being disabled. A harvester requires a longer period to recuperate after being disabled.

- Once recovered, the onboard power system of the affected rover or harvester is immune from future shots of the opponent's rover(s) which caused it to overload. A rover may however still be disabled again by other rovers from the opposing team who have yet to shoot it.

- Rovers may help accelerate the recovery of their harvester by transmitting repair codes to their harvester. Repair codes can be obtained by firing a laser-coded message to the processing plant (refer to your RCLS manual for more information).

A pulse-coded laser system is made up of a pulse-coded **laser AND **detector and is connected to the RCLS of your rover/harvester**.*

Technical Requirements:

- **Rover/Harvester**: The centre of the laser detector must be placed vertically at an exact height of 10cm from the playing surface without any obstructions. If in doubt, ask the instructors about the placement as incorrect placement can result in disqualification.

Procedure:

- Rover uses its pulse-coded laser system to fire a shot at either the opponent's rovers or harvester.
- Successful hit of an opponent's rover disables it for a fixed period.
- 3 successful hits of an opponent's harvester will disable it for a fixed period (which is longer than the time for a disabled rover).
- Once recovered, the rover or harvester regains normal functionality.
- Rovers can obtain repair codes from the processing plant and transmit them to their harvester to help them recover once they have been hit.

Rules and Constraints:

- All rovers and harvester **must be equipped** with a (**provided**) pulse-coded laser system.
- A rover can only disable a specific opponent's rover once.
- A rover can only hit a harvester once. A harvester will only be disabled after 3 hits.
- A harvester does not need to be disabled to receive a repair code but can only receive one after it has been hit at least once.
- Receiving a repair code when not yet disabled will remove a hit from the harvester.
- Receiving a repair code when already disabled will reduce the disabled period of the harvester.

- A rover can only obtain one repair code from the processing plant.

Tips and Strategies:

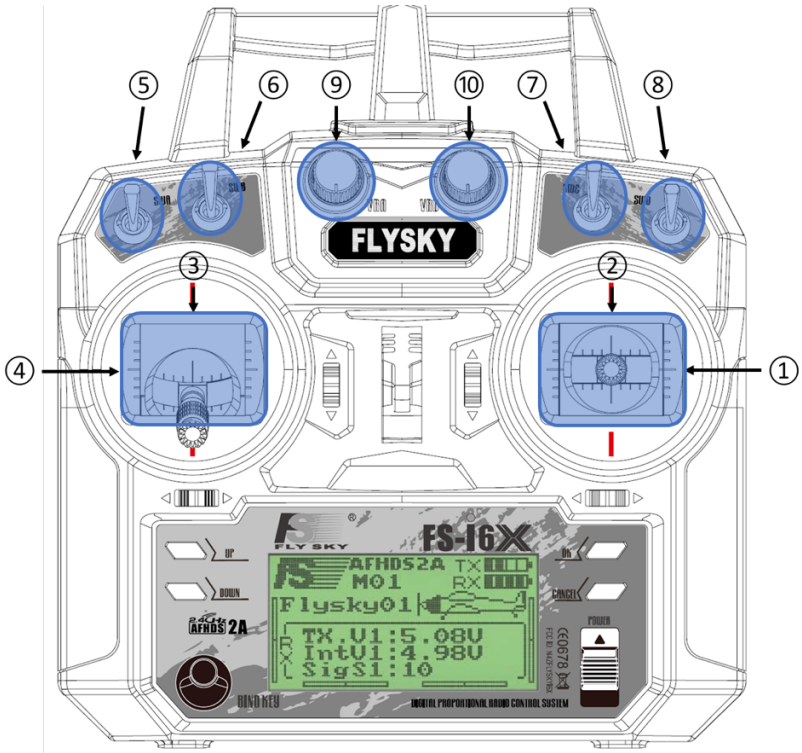
- Strategically plan when you want to attempt to disable their harvester by not allowing the opponent's rovers time to transmit repair codes during your assault.
- Coordinate the rovers' access and delivery of repair codes for the harvester such that it is not disabled or to reduce the disabled period during the round.

TECHNICAL SPECIFICATIONS

In-game user control of Harvester/Rover

Overview:

Although multiple students are working on each harvester or rover, only one player can interact with a specific vehicle during a competition round. This implies that each team will have 5 operators during a competition round (i.e. 1 for the team's harvester and 1 per rover). Each team will use five 10 channel wireless transmitters for the competition. The transmitters consist of both left and right 2-axis joysticks, three 2-position toggle switches, one 3-position toggle switch, and 2 variable potentiometer dials. It will be up to each group to decide how these 10 channels will be used to control the corresponding harvester/rover.



① Right Joystick X-Axis
[Range: 1000-2000,Returns to Centre]

② Right Joystick Y-Axis
[Range: 1000-2000,Returns to Centre]

③ Left Joystick Y-Axis
[Range: 1000-2000]

④ Left Joystick X-Axis
[Range: 1000-2000,Returns to Centre]

⑤ Switch A (SWA)
[Values: 1000 or 2000]

⑥ Switch B (SWB)
[Values: 1000 or 2000]

⑦ Switch C (SWC)
[Values: 1000, 1500, or 2000]

⑧ Switch D (SWD)
[Values: 1000 or 2000]

⑨ Dial A (VRA)
[Range: 1000-2000]

⑩ Dial B (VRB)
[Range: 1000-2000]

PROCESSING PLANT

Overview

Each team will have a processing plant at their home base. Aside from being the location where ore is brought back and dispensed for maximum points, this plant will also be involved in receiving repair code requests through its equipped laser detector and RCLS.
A rectangular ore hopper will be attached to the front face of the processing plant as shown below.

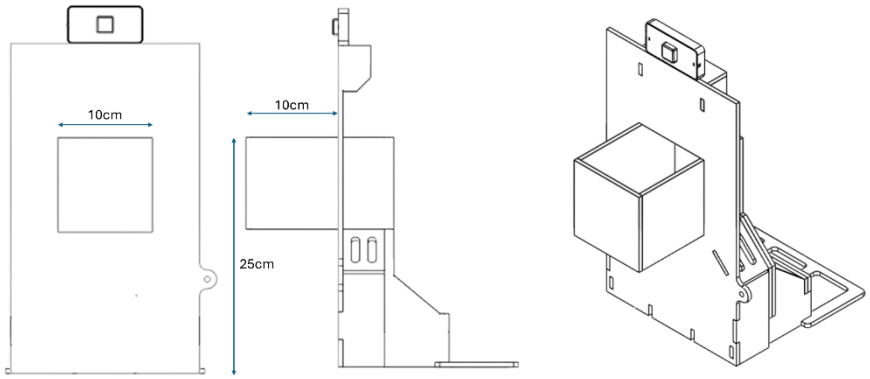


Figure 14 Processing Plant and Hopper

GATES

Overview

Nearly every task will have a gate system associated with it that will unlock access to an area with ore and/or unlock points. The gate systems will be each be controlled by a BBC micro:bit and breadboard system as shown below in Figure 15 BBC Micro:bit Figure 15. Additionally, there will be a “traffic light” system associated with each task to indicate if it is safe to enter and retrieve the ore and if points have been awarded.

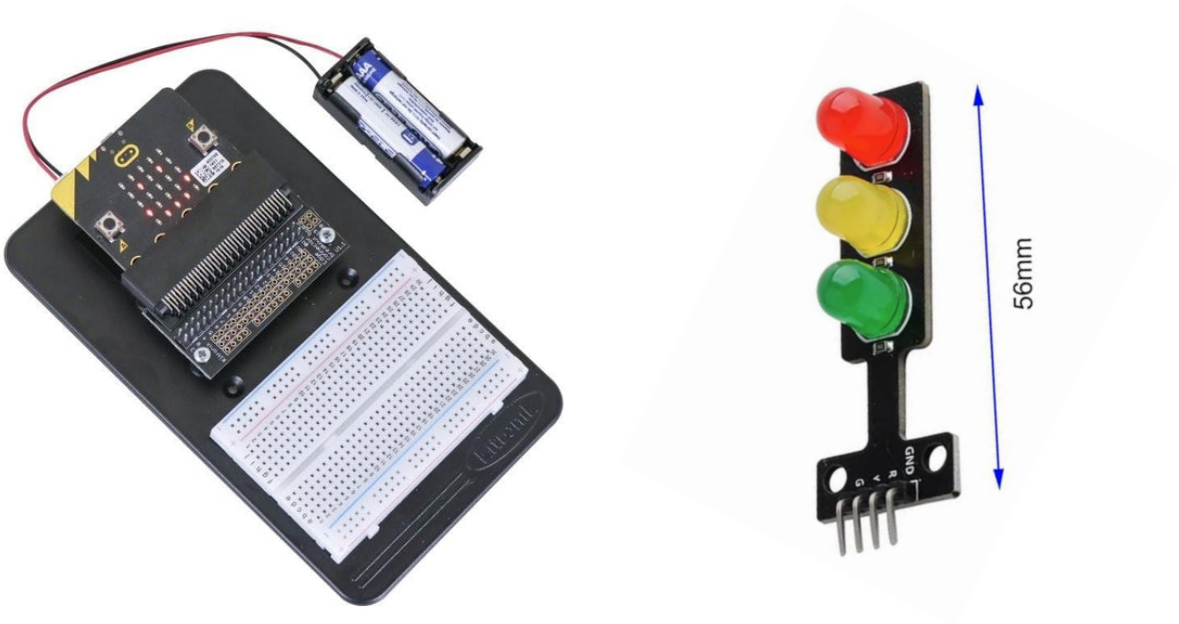


Figure 15 BBC Micro:bit, breadboard and light system

When the red LED on the traffic light is lit up, rovers and harvesters may NOT enter the area the light is in front of. Once the task has been completed, the red LED will turn off and the green LED will turn on, indicating the area is now available for ore retrieval and points have been awarded.

ORE

Overview

Ore is king in this competition, the more you have, the more points your team will acquire. However, it's not as simple as just collecting as much ore as you like. There are 3 primary types of ore and 2 bi-stable types. Primary ore may NOT be mixed together whereas as bi-stable may be mixed with a corresponding primary only. See Figure 16 Ore Types below.

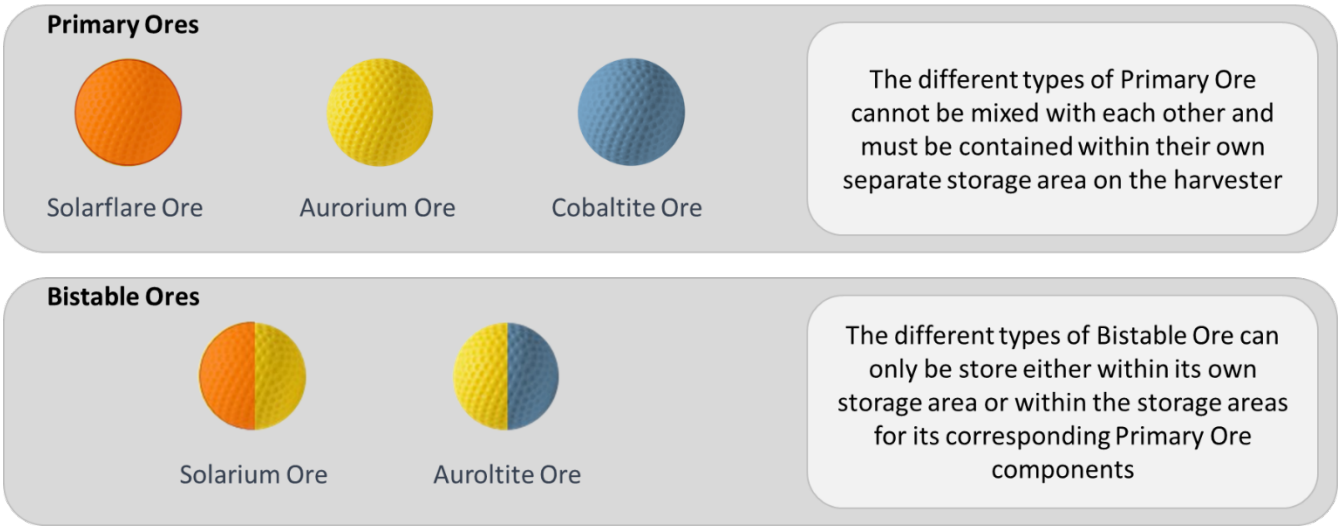


Figure 16 Ore Types

TEAM STRATEGY AND PLANNING

Overview

We've done our best to provide only the necessary constraints to this competition to allow for freedom in design and strategy. While it's good to check in with the course instructors with questions you have, take note that we've also intentionally left aspects of the rules open to allow flexibility and variation in designs.

Take time to consider what each group's role will be in the competition. Some strategies may have too high of a risk/reward whereas others may be so quick to complete that they are low hanging fruit. We encourage creativity in your strategy and hope you spend time working with your team to develop it.

Please remember to keep checking back in this manual for updates and we will do our best to keep you informed. The FAQ section will be updated as we get (and answer) questions.

FAQ

Question: